Advanced Programming Practice · 21CSC203P

Unit 4 · Session 8 · SLO 2 — Solved Worksheet

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Unit â€“ 4 : Session â€“ 8 : SLO â€“ 2 | Shared vs Distributed Models â€” Real-World Application

I. Discussion Question

• Application: Google Search (large-scale web search engine).

• Why distributed computing: the web index is petabytes in size â€” far beyond any single machine â€” so it is sharded across thousands of servers, each searching its own partition in parallel.

• Scalability: more machines are simply added as query volume and index size grow (horizontal scaling).

• Fault tolerance: data and services are replicated, so individual machine failures do not interrupt search results.

• Low response time: partial results from many nodes are aggregated in parallel, returning answers in milliseconds.

• Other valid examples: Hadoop/MapReduce batch analytics, Netflix video streaming (CDN), blockchain networks, weather simulation clusters.